

Anti-Disturbance Self-Supervised Reinforcement Learning for Perturbed Car-Following System

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Abstract—This paper proposes an anti-disturbance car-following strategy for attenuating (i) exogenous disturbances from preceding traffic oscillations and (ii) endogenous disturbances in vehicular control systems (e.g., wind gust, ground friction, and rolling resistance). Firstly, it employs a modified robust controller to generate an expert car-following control experience. Subsequently, it imitates the expert behaviors via the behavioral cloning (BC) technique, thereby developing the anti-disturbance ability. Lastly, the obtained policy is optimized using the self-supervised reinforcement learning (RL) approach. The simulation experiments, comprising both training and evaluation phases, are performed via Python. To simulate car-following scenarios, we utilize the ground-truth data from the Next Generation Simulation (NGSIM) datasets. Through recursive interactions with the perturbed car-following environment, self-supervised RL drives stable policy improvement. The proposed anti-disturbance self-supervised RL (ADSSRL) policy presents a smooth and almost monotonously increasing reward curve. Further evaluation of disturbance dampening performance suggests that at least a 44.5% reduction in control efficiency cost and a 10.1% reduction in driving comfort cost are achieved compared with baselines.

Index Terms—Anti-disturbance, car-following, self-supervised reinforcement learning, traffic oscillation.

I. INTRODUCTION

CONNECTED autonomous vehicles (CAV), equipped with advanced communication and automation capability, have the potential to improve traffic capacity, reduce crash risks, and improve driving comfort [1], [2], [3]. As one of the most elementary driving tasks, CAV car-following control usually suffers from two kinds of disturbances. One is exogenous disturbances from preceding traffic oscillations. Speed disturbances might be amplified in a vehicle platoon, resulting in low fuel efficiency, increased crash likelihood, and severe traffic congestion [4], [5], [6]. The other is endogenous disturbances originating from wind gust, ground friction, rolling resistance, and uncertainty of vehicular parameters [7]. These factors make CAV control sluggish, unstable, or even error-prone because precise vehicular

movement control (e.g., achieving expected acceleration) is affected.

Numerous previous studies have tried to resolve these disturbances using linear or nonlinear approaches [7], [8], [9], [10]. Many scholars designed adaptive control law to specially handle parameter uncertainties in vehicular control systems [3], [11], [12]. Later, several H-infinite controllers were proposed to achieve accurate acceleration tracking considering more complex endogenous disturbance situations [10], [13]. Further, recent studies designed robust feedback and feedforward controllers to reduce the difference between desired and actual accelerations under perturbations [7], [14]. They also proved the robust stability of a vehicle platoon against various disturbances. However, the abovementioned approaches have not explicitly formulated collision-free constraints within acceptable acceleration/deceleration limits. Additionally, it is challenging to design a multi-objective optimization framework to improve the car-following performance.

Considering the above limitations, model predictive control (MPC)-based approaches were applied. They integrated parametric estimation and stochastic optimization into MPC to improve the robustness of the car-following system [15], [16], [17]. However, the above models were built based on a normally distributed disturbance. To relax the probability distributed-related assumption for endogenous disturbances, a widely-used robust MPC was developed, the so-called min-max MPC [18], [19], [20], [21]. It was designed to optimize the worst-case model performance considering all constraints. These studies achieved effective and practical vehicle control with mixed stochastic and deterministic disturbances. However, the MPC-based control scheme usually requires the problem to be convex so that it can be solved efficiently [22]. In addition, vehicle-to-vehicle (V2V) wireless communication conditions must be satisfied to obtain the preceding vehicle's control actions in these studies. When encountering communication delay or failure, the approaches may not work effectively.

Recently, deep reinforcement learning (DRL) algorithms have been successfully applied in various vehicular control tasks, including lane keeping, lane changing, and eco-driving [23], [24], [25], [26], [27]. By a trial-and-error way, the DRL agents can learn and explore optimal actions from past control experiences. DRL policies achieve acceptable control performance by capturing complex and stochastic system characteristics, even without V2V communication conditions [22]. Hence, DRL is a promising approach to obtain optimal solutions from perturbed car-following environments. Several studies adopted

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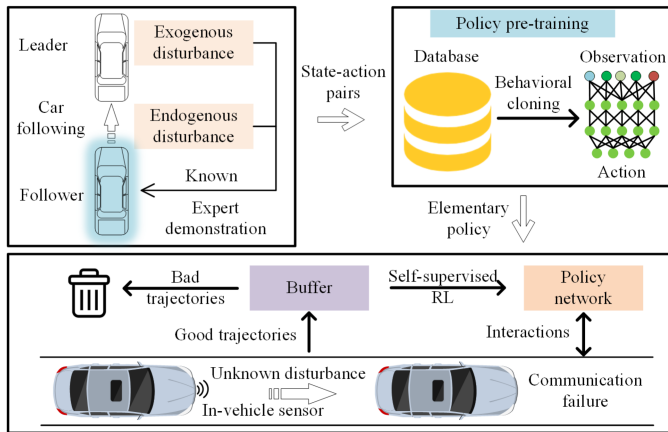


Fig. 1. Control framework of the ADSSRL-based strategy.

single-agent DRL to alleviate traffic oscillations with certain goals (e.g., low energy consumption) [28], [29], [30], [31], [32]. They demonstrated that well-trained DRL agents could effectively dampen traffic oscillations with high energy efficiency [28], [29]. More research sought cooperative CAV control to generate safe, efficient, and comfortable speed profiles. Shi et al. [22] adopted the distributed single-agent DRL to attain multi-objective optimization, including driving efficiency, safety, fuel consumption, and platoon stability. They also extended the DRL framework in mixed human-driven vehicle and CAV traffic with communication failure [33]. Later, Li et al. [34] developed one state and one reward channel to provide abundant CAV coordination information. It demonstrated that vehicles controlled by several parallel agents could achieve a collective goal in traffic oscillations. More recent research by Jiang et al. [4] found that a cooperative DRL-based controller could generate smaller oscillation growth, overshooting, and average acceleration/deceleration.

However, most current DRL-based car-following researches focus on exogenous disturbances while endogenous disturbances are neglected, e.g., [4], [28], [34]. In addition, traditional DRL algorithms are reported to be notoriously unstable due to the high variance of policy gradient [35], [36], [37], [38]. DRL policy tends to be brittle or sensitive when receiving spurious rewards induced by unobserved disturbances in the training stage [39]. Therefore, it cannot learn a well-satisfied policy if simply applying DRL to the mitigation of two kinds of disturbances at the same time.

In this article, we propose an anti-disturbance self-supervised RL (ADSSRL) algorithm for vehicle control in perturbed car-following situations. Anti-disturbance refers to the ability of a control system to reject or be robust to external disturbances, thus enabling it to maintain the desired performance in the presence of disturbances. This work conducts a pilot study applying DRL to overcome exogenous and endogenous disturbances simultaneously. Since supervised regression has been demonstrated to be a more stable way to train deep policy networks [40], [41], we adopt supervised learning to train DRL agents in the perturbed environment. The primary contributions of this article are as follows.

- 1) This study integrates the merits of the supervised learning algorithm (stable training) and RL algorithm (working beyond human knowledge) to drive stable policy improvement for perturbed car-following control.
- 2) Instead of randomly searching for the optimal policy, a well-designed robust car-following controller is employed to guide policy learning, which speeds up RL training and empowers the anti-disturbance ability to RL agents.

The rest of this paper is organized as follows. Section II provides the model framework and car-following system. Section III presents model details and training environment. Section IV illustrates experiments and results. Conclusion and discussion are shown in Section V.

II. CONTROL SYSTEM FRAMEWORK

A. Model Framework

This study aims at generating an anti-disturbance car-following policy using supervised learning and RL. The proposed ADSSRL-based strategy contains three sequential steps, as illustrated in Fig. 1. Firstly, we introduce the concept of ‘expert demonstration’. It is a feasible solution to a challenging problem enabled by current human knowledge [42], [43]. It can facilitate effective exploration and thus speed up RL training in uncertain environments. Since robust control techniques typically guarantee robustness under system perturbations, we employ one, i.e., generalized extended state observer-based control (GESOBC) [8], to control a series of car-following behavior. It generates an expert demonstration. We alter the controller by giving exogenous and endogenous perturbations to further enhance the performance of the expert demonstration.

Subsequently, we decompose the expert demonstration into a series of car-following state-action pairs in a discrete fashion. All state-action pairs are stored in an RL’s replay buffer. A behavioral cloning (BC) technique is utilized to fit the sampled data from inputs (i.e., observations) to outputs (i.e., actions). In this way, it learns an elementary policy by imitating the behaviors of expert demonstration in the pretraining stage. Note that the state and action are defined in the RL formulation section.

Thirdly, the RL agent recursively interacts with the perturbed car-following environment to obtain various control experiences. High-quality control experiences are selectively preserved in the replay buffer, while inadequate ones are discarded. This process leads to a gradual improvement in the overall data quality of the replay buffer. The previously generated elementary strategy is thus updated through self-supervised RL using improved training data until convergence.

Notably, steps one and two are executed only once, while the third step is performed iteratively. Learning policy from an existing robust controller can speed up training and empower the anti-disturbance ability to DRL agents in the perturbed car-following environment. The generated expert demonstration remains in the replay buffer almost throughout the training period, so it can always guide RL policy training. The self-supervised RL algorithm facilitates the training approach. In this way, the ADSSRL agent gradually learns better policies and eventually

surpasses the expert demonstration through interactions with random traffic environments.

In a car-following system, the V2V communication technology usually enables CAVs to react to downstream speed disturbances in advance and thus achieve better car-following performance in safety, mobility, and sustainability. However, the success rate of V2V communication is usually influenced by the fading effect of the signal amplitude over distance, particularly when vehicles simultaneously exchange and disseminate data [33]. The widely-adopted IEEE 802.11p communication protocol also usually suffers from significant packet reception degradation due to communication interference by transmissions from nearby vehicles [44]. Hence, our proposed strategy is developed in the communication failure situation. All required information in this study is obtained from onboard sensors instead of V2V communication, making the proposed strategy more flexibly implemented with less communication burden.

B. Perturbed Car-Following System

The perturbed car-following system is presented in this subsection. For a car-following system, there is a follower (vehicle i) and a leader (vehicle $i-1$). Vehicle i must be an automated vehicle to be controlled so that we can change its driving state and car-following state. The vehicle car-following process usually has an expected state. It generally includes three objectives: (i) regulating the inter-vehicle spacing according to a certain (equilibrium) spacing policy; (ii) maintaining zero speed difference with the preceding vehicle; (iii) guaranteeing driving safety. According to the Society of Automotive Engineer Standard, a constant time gap (CTG) policy is adopted in this study as it can guarantee traffic safety, improve string stability, and ensure driving comfort. CTG describes the desired spacing as

$$d_i^*(t) = v_i(t) * \tau_i^* + l_i \quad (1)$$

where $d_i^*(t)$ is the desired spacing of vehicle i at time t . $v_i(t)$ is the instant speed of vehicle i at time t . τ_i^* and l_i are the expected time gap and standstill spacing of vehicle i , respectively.

We define the deviation from desired spacing as $\Delta d_i(t)$ and the speed difference between vehicle i and its leader (i.e., vehicle $i-1$) as $\Delta v_i(t)$. They are calculated as follows.

$$\Delta d_i(t) = d_i(t) - (v_i(t) * \tau_i^* + l_i) \quad (2)$$

$$\Delta v_i(t) = v_{i-1}(t) - v_i(t) \quad (3)$$

where $d_i(t)$ is the actual spacing of vehicle i at time t ; $v_{i-1}(t)$ is the speed of vehicle $i-1$ at time t . They describe the error with the expected car-following states. An effective control strategy should reach or maintain the equilibrium state with $\Delta d_i(t) = 0$ and $\Delta v_i(t) = 0$ by adjusting vehicle accelerations.

In practice, the desired acceleration cannot be achieved promptly due to the actuation time lag. The realistic acceleration is calculated as in (4), considering actuation time lag and endogenous disturbances.

$$\dot{a}_i(t) = -\frac{1}{T_i} a_i(t) + \frac{\bar{K}_i}{T_i} u_i(t) + \delta_i(t) \quad (4)$$

where $a_i(t)$ is the acceleration of vehicle i at time t . $\dot{a}_i(t)$ is the derivative of acceleration. $u_i(t)$ is the desired acceleration. T_i is the actuation time lag to achieve the required acceleration. $\bar{K}_i$ is the ratio of the required acceleration that can be achieved. $\delta_i(t)$ contains all possible disturbances in the vehicular control system. $\bar{T}_i$ and $\bar{K}_i$ are the presetting constant parameters representing the average value of T_i and K_i , respectively.

We define $x_i(t) = [\Delta d_i(t), \Delta v_i(t), a_i(t)]^T$ as the system state and $u_i(t)$ as the system input. Then, we have the state-space system as follows.

$$\dot{x}_i(t) = A_c x_i(t) + B_c u_i(t) + D_c \begin{bmatrix} a_{i-1}(t) \\ \delta_i(t) \end{bmatrix} \quad (5)$$

$$\text{where } A_c = \begin{pmatrix} 0 & 1 & -\tau_i^* \\ 0 & 0 & -1 \\ 0 & 0 & -\frac{1}{T_i} \end{pmatrix}, B_c = \begin{pmatrix} 0 \\ 0 \\ \frac{\bar{K}_i}{T_i} \end{pmatrix}, D_c = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

In (5) exhibits how the control input changes the car-following states. When a vehicle suffers from V2V wireless communication failures, its preceding vehicle's acceleration, $a_{i-1}(t)$ is unavailable. Therefore, one can find two disturbance factors influencing the control system. Specifically, $a_{i-1}(t)$ reflects the exogenous disturbances originating from traffic oscillations, and $\delta_i(t)$ reflects the endogenous disturbances in the vehicular control system. According to Chen et al. [7], the endogenous disturbances mainly come from wind gusts, rolling resistance, friction on the grounds, engine time constant, and the mass of passengers. They are composed of $\delta_i(t)$ as an unknown but bounded disturbance value.

III. METHODOLOGY

A. RL Formulation

We use the Markov game to model the ADSSRL process, typically formulated as a Markov decision process (MDP). It can be formalized by a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma, p_0)$, where $\mathcal{S}$ is the continuous state space in the car-following system, $\mathcal{A}$ is the continuous action space, $\mathcal{P} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{S}$ is the state transition probability, r is the reward function, $\mathcal{R}$ is the reward space, $\gamma \in (0, 1)$ is the reward discount factor, and p_0 is the distribution of the initial state. Each agent i produces an action a_i based on the perceived state s_i . After the implementation of actions, it receives the next state and reward feedback, r_i . The crucial components: state, action, and reward, are defined as follows.

1) *State Representation*: To incorporate the car-following system, three state variables in the state-space system (5) are integrated into the state vector of RL. Specifically, we define the state vector $\mathbf{S} = [\Delta d_i(t), \Delta v_i(t), a_i(t)]$. They represent the spacing deviation, speed deviation, and acceleration deviation from the equilibrium point $\mathbf{S}_0 = [0, 0, 0]$, respectively.

2) *Action Selection*: As a decision-making problem, the control input in the car-following system is set as the action variable of RL. It indicates the desired acceleration to be achieved. According to previous DRL-based vehicular control studies, the desired acceleration is set as a continuous variable within $(-3 \text{ m/s}^2, 3 \text{ m/s}^2)$ to maintain a comfortable driving condition [34], [45].

3) *Reward Design*: The reward in this study consists of three parts: control efficiency, driving comfort, and traffic safety. The control efficiency f_i^t measures the cost of deviation from the equilibrium spacing ($\Delta d_i(t)$) and speed deviation ($\Delta v_i(t)$) in a quadratic form as follows.

$$f_i^t = [\Delta d_i(t), \Delta v_i(t)] Q_i [\Delta d_i(t), \Delta v_i(t)]^T \quad (6)$$

where Q_i is a positive definite diagonal coefficient matrix with two weights α_1, α_2 , defined as $Q_i = \begin{pmatrix} \alpha_1 & 0 \\ 0 & \alpha_2 \end{pmatrix}$. The driving comfort term gives $g_i^t = R_i (a_i(t))^2$ where R_i is the acceleration weighting coefficient.

We adopt traffic conflict to represent the traffic safety state in the car-following process. Traffic conflict is a safety index to represent the potential of crash risks. A traffic collision will likely occur when the traffic conflict index exceeds the predefined threshold. It can achieve more rigorous safety guidance. In practice, the traffic conflict index informs the subject vehicle to decelerate proactively to avoid crashes. The deceleration rate to avoid a crash (DRAC) is a commonly used traffic conflict index in many previous studies, especially in evaluating rear-end crash risks [34], [46]. It is defined as the minimum deceleration rate required by the following vehicle to match the velocity of the leading vehicle to avoid a crash, as written in (7).

$$DRAC_i(t) = \begin{cases} \frac{(v_{i-1}(t) - v_i(t))^2}{d_i(t) - L_{i-1}}, & \text{if } v_i(t) > v_{i-1}(t) \\ 0, & \text{else} \end{cases} \quad (7)$$

where L_{i-1} is the vehicle length of the vehicle $i-1$.

A threshold index, maximum available deceleration rate (MADR) is applied to judge whether a traffic conflict occurs. One vehicle is deemed to have a probability of colliding with the leading vehicle if $DRAC_i(t) > \text{MADR}$. A widely used MADR value, 3.4 m/s^2 is adopted in this study [34], [46]. Based on that, we can obtain the safety reward value $r_{s,i}^t = -1$ if $DRAC_i(t) > \text{MADR}$. Otherwise, $r_{s,i}^t = 0$. We adopt the total reward function specified in (8).

$$r_{total} = r_{s,i}^t + \exp(-f_i^t - g_i^t) \quad (8)$$

In the reward function, $r_{s,i}^t$ represents traffic safety while $\exp(-f_i^t - g_i^t)$ denotes the costs associated with control efficiency and driving comfort. The equation is carefully designed to make sure that traffic safety is a dominant term. Specifically, the total reward is nonnegative when there is no traffic conflict, whereas it becomes negative in the presence of a traffic conflict. This negative reward prompts vehicles to avoid dangerous driving conditions. Moreover, the inclusion of control efficiency and driving comfort terms ensures the feasibility and adaptability of the control strategy. The design drives the car-following system to the expected equilibrium states, $\mathcal{S}_0$ with acceptable performance and fast real-time computation.

B. ADSSRL-Based Car-Following Strategy

In this section, we present how the ADSSRL policy π_θ updates in the training loop. The core idea is to search for better-performed behaviors from a new policy π'_θ than those from the current policy π_θ . Firstly, the policy π_θ interacts with

Algorithm 1: ADSSRL-Based Car-Following Algorithm.

Input: Training steps $\mathcal{S}$, buffer $\mathcal{D}$

Run GESOBC with known disturbances to generate a car-following control experience ξ

Store all state-action pairs in ξ into buffer $\mathcal{D}$

Pre-train an elementary policy π_θ through minimizing the regression loss as in (9)

For episode = 1, 2, ... until converge **do**

Execute π_θ to generate a series of car-following state-action pairs (s_i, a_i) within episode J

Assign the episodic reward R_J to all (s_i, a_i) and store them in buffer $\mathcal{D}$

Rank the state-action pairs in $\mathcal{D}$ based on reward and discard the ones with low reward

For step = 1, 2, ..., $\mathcal{S}$ **do**

Sample a batch of state-action pairs from $\mathcal{D}$ and update π_θ with (9)

End for

End for

the car-following environment. After an episode, we can obtain a series of state-action pairs. It assigns the episodic reward R_J (i.e., the sum of all reward values over the episode) to all pairs within the episode and stores them into the buffer. After several episodes, the ranking buffer selects highly rewarded state-action pairs by discarding the pairs with low episodic rewards. It remains a fixed number of state-action pairs in the buffer. Then, BC is adopted to improve the policy π_θ via minimizing the mean-square error to sampled data. Specifically, for a sample batch $\{(s_k, a_k)\}_{k=1}^N$, the policy is updated by (9).

$$\min L = \frac{1}{N} \sum_{k=1}^N (\pi_\theta(a_k|s_k) - a_k)^2 \quad (9)$$

where N is the batch size; $\pi_\theta(a_k|s_k)$ is the action produced by policy π_θ given state s_k .

The π_θ is a stochastic policy that samples actions from Gaussian distributions. It enables the RL agent to reach many valuable trajectories due to the occasional randomness in the car-following control system. RL defines ‘good’ experiences as those that result in high episodic rewards. This is because episodic rewards are commonly used as the metric for evaluating the performance of an agent. As a result, the state-action pairs obtained from high-rewarded episodes are considered to be ‘good’ experiences and are more appropriate for expert demonstrations. Conversely, the state-action pairs from low-rewarded episodes are regarded as ‘bad’ experiences in this study. We discover that imitating ‘bad’ driving behaviors during training can actually misdirect policy training, rendering it challenging to attain convergence. As a result, we remove these experiences from the replay buffer to guarantee that agents concentrate on the ‘good’ experiences that lead to high rewards. Consequently, the RL policy is updated by imitating actions from the replay buffer via BC in (9). Note that the notation a_i representing RL action is different from the notation of acceleration $a_i(t)$. We

summarize the algorithm of the ADSSRL-based car-following control strategy in Algorithm 1.

The algorithms are residing on an onboard computer to control the car-following systems. The onboard computer is typically located in the vehicle's dashboard or center console and is responsible for managing various vehicle systems. By deploying our algorithm on this computer, it aims to enhance the vehicle's car-following capabilities, thereby improving its safety and efficiency.

C. Baselines for Comparison

We introduce three baseline strategies for comparison. Two are traditional RL-based car-following controllers, and the third is a robust controller in literature.

1) *DDPG-Based Car-Following Control Strategy*: As a widely-used RL algorithm in continuous action space, Deep Deterministic Policy Gradient (DDPG) has been successfully applied in the microscopic vehicle control field [30], [45]. This study uses DDPG as a baseline to train agents in the perturbed car-following system.

2) *PPO-Based Car-Following Control Strategy*: Proximal Policy Optimization (PPO) is a state-of-the-art on-policy RL algorithm [47]. It has a more stable training performance than DDPG by limiting the new policy and not getting far from the old policy using a Kullback-Leibler (KL) divergence constraint. It is also a widely-used RL algorithm in longitudinal vehicular control [22], [33].

3) *GESOBC-Based Car-Following Control Strategy*: Chen et al. [7] designed a robust controller to guarantee string stability for vehicle platoons with communication failure and lumped disturbances. The GESOBC control law is represented in (10).

$$u_i(t) = k_{xi} x_i(t) + k_{di} \begin{bmatrix} \hat{a}_{i-1}(t) \\ \hat{\delta}_i(t) \end{bmatrix} \quad (10)$$

where k_{xi} denotes the feedback control gains and k_{di} denotes the disturbance compensation gains. $\hat{a}_{i-1}(t)$ and $\hat{\delta}_i(t)$ are the estimation of the corresponding variables. It addressed the issue with nonzero initial conditions considering the communication failure situations and random disturbances. Through pole placement, we can obtain the corresponding parameters. Readers can refer to Chen et al. [7] for more details. Notably, the DDPG and GESOBC are two separate control algorithms. The DDPG policy employs two distinct actor and critic networks to explore and learn the desired accelerations. In contrast, the GESOBC policy is a well-designed controller developed using control theory, which satisfies the theoretical string stability of a vehicle platoon.

D. Training Environment

In this section, we give the training procedure and parameters. The training environment is built via Python. The trajectory of the leading vehicle is ground-truth data from the Next Generation Simulation (NGSIM) datasets of I-80 in California, including the typical feature of acceleration, deceleration, and uniform speed. It enables the proposed control strategy to consider the characteristic of empirical traffic data. The raw field data was

filtered and reconstructed by [48] and [49] using a low-pass filter to eliminate the data noises. Vehicle trajectories from 4:00 pm to 4:15 pm in Lane 2 are adopted due to the frequent traffic oscillations during this time. Without loss of generality, all vehicles are initialized with the equilibrium states defined. The exogenous disturbance for the 1st agent is the acceleration of the platoon's leading vehicle. This acceleration is obtained from empirical data from the NGSIM datasets. The trajectory of the leading vehicle from the NGSIM data enhances the practicality of our approach for real-world scenarios.

Notably, during the training of ADSSRL policy, adjusting the parameters k_{xi} and k_{di} is possible, but it is a time-consuming task that requires a complex tuning approach to ensure the stability of the system under various conditions. To enhance the efficiency of the RL policy training, we have incorporated the well-designed control gains from the GESOBC policy. The two control gains can be set as the default values. Based on that, model optimization can be executed, considering the default control gains as the starting point. We only consider reward design and model structure for algorithm optimization, and more analysis for control gains will be explored in future researches.

The safety implications of an experimentally tested approach should be considered. It depends on various factors such as the specific design of the RL algorithm, the level of autonomy of the vehicles, and the conditions under which the experiment is conducted. In fact, our proposed approach has paid great attention to safety performance. For model development, the ground-truth datasets have been embedded in training to better incorporate stochastic characteristics of human-driven vehicles into control. Incorporating the randomness of real-world traffic makes the strategy more applicable to field testing. Subsequently, traffic safety has been set as the dominant term in the reward function. In other words, the proposed strategy regarded traffic safety as the most significant control objective. Moreover, the safety implications of an RL-based car-following approach would also depend on the level of autonomy of the vehicles. If the vehicles are fully autonomous, the safety implications would be greater since there would be no human driver to take control in case of a malfunction or emergency. On the other hand, if the vehicles are only partially autonomous, the safety implications would be reduced since a human driver would still be able to intervene if necessary.

The ADSSRL-based strategy aims at maintaining the car-following equilibrium states while dampening the preceding speed disturbances. The training lasts for 1000 episodes for all RL algorithms, consisting of 500 timesteps with a 0.1s time interval. The main control parameters in the simulations are presented in Table I.

IV. EXPERIMENT RESULTS

A. Training Results

The ADSSRL-based strategy is trained through iterative interactions with traffic environments. The DDPG and PPO algorithms are also applied for comparison. Different values are tried to search for the most suitable hyperparameters for convergence, and the final parameters are presented in Table II. Rewards

TABLE I
PARAMETER VALUES IN SIMULATION

Parameter	Description	Value
τ_1^*	Expected time gap	1s
sd_1	Standstill spacing	7m
L_0	Vehicle length	5m
Q_i	Weighting matrix of control efficiency	$\begin{pmatrix} 1 & 0 \\ 0 & 0.5 \end{pmatrix}$
R_i	Weighting matrix of driving comfort	0.5
$\bar{T}_i$	Actuator lag to achieve the required acceleration	0.25
$\bar{K}_i$	The ratio of the required acceleration implementation	0.75
k_{xi}	Feedback control gains	[0.78, 0.91, -0.05]
k_{di}	Disturbance compensation gains	[0.3165, -0.2941]

TABLE II
MAIN HYPER-PARAMETERS IN RL TRAINING

Hyper-parameters	DDPG	PPO	ADSSRL
Learning rate	0.001	0.0001	0.01
Discount factor	0.90	0.90	None
Memory capacity	80000	None	2000
Batch size	512	512	512
Soft replacement	0.01	None	None
Updating bound	None	0.2	None

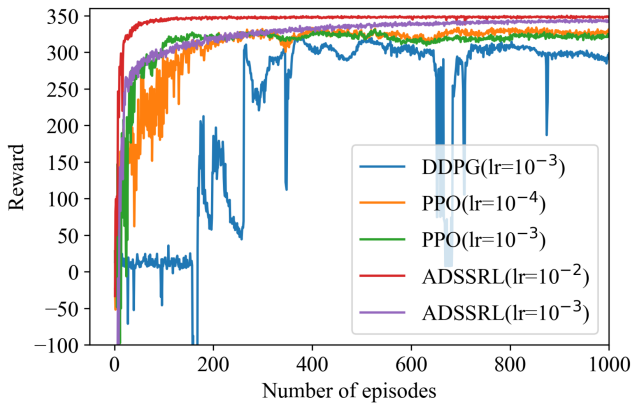


Fig. 2. Reward curves of different RL-based approaches.

against the number of training episodes are illustrated in Fig. 2. Several values (0.1, 0.01, 0.001, 0.0001) were tested as hyperparameters in order to determine the most suitable parameters for achieving convergence. The findings indicate that the ADSSRL policy is highly robust to changes in learning rate. Specifically, convergence can be achieved when the learning rate is no more than 0.01. In contrast, the DDPG and PPO algorithms can only converge when the learning rate is no more than 0.001. The reason for this discrepancy is that the supervised regression to expert demonstration approach used in ADSSRL is more stable than traditional RL using the policy gradient. This property enables ADSSRL to utilize a larger learning rate, leading to faster convergence.

We also compare the training performance of different RL algorithms utilizing the same learning rate. Three algorithms

were trained using a learning rate of 0.001, and the reward curves were presented in Fig. 2. The results indicate that our proposed ADSSRL algorithm still achieves a greater reward compared to the DDPG and PPO algorithms. A higher episodic reward reflects that the agent has learnt a better strategy. It is also found that the same algorithm with different learning rates yields similar reward values, as the learning rate primarily impacts the speed of training.

One can notice that the first episodic reward appears to be similar across these RL methods. It is because of the gradual learning process. Specifically, during the initial training stage, the RL agents have not yet updated their neural networks and instead adopt random actions. As a result, the random actions taken during this stage lead to an approximation of the episodic reward values in the first episode.

Here we analyze the reason for stable training of the proposed strategy in the perturbed car-following system as follows. Conventional RL algorithms update policy relying heavily on the instantaneous reward that reflects the control effects at that moment. Therefore, the unstable reward feedback induced by random disturbances may misguide RL training. Conversely, the ADSSRL updates the policy by optimizing the episodic reward. It describes the aggregated rewards over a whole episode, which is insensitive to the fluctuation of the instantaneous reward. Hence, the ADSSRL framework can overcome the brittle drawbacks of traditional RL in perturbed environments. In addition, training through minimizing the regression loss is more stable than the policy gradient. Thereby, the proposed strategy gains the ability of anti-disturbances.

B. Policy Evolution During Training

To understand how the ADSSRL-based car-following strategy behaves during policy learning, we visualize the control input and car-following states during different training stages. The training performance after pretraining, training for 10 episodes, training for 100 episodes, and training for 1000 episodes are presented in Fig. 3. In order to evaluate the control performance of the proposed strategy, two representative indices are used, i.e., the control efficiency cost $\bar{f}_i$ and the driving comfort cost $\bar{g}_i$ as in [22], [33]. They are the average value of f_i^t and g_i^t throughout an episode. The former indicates the performance of dampening

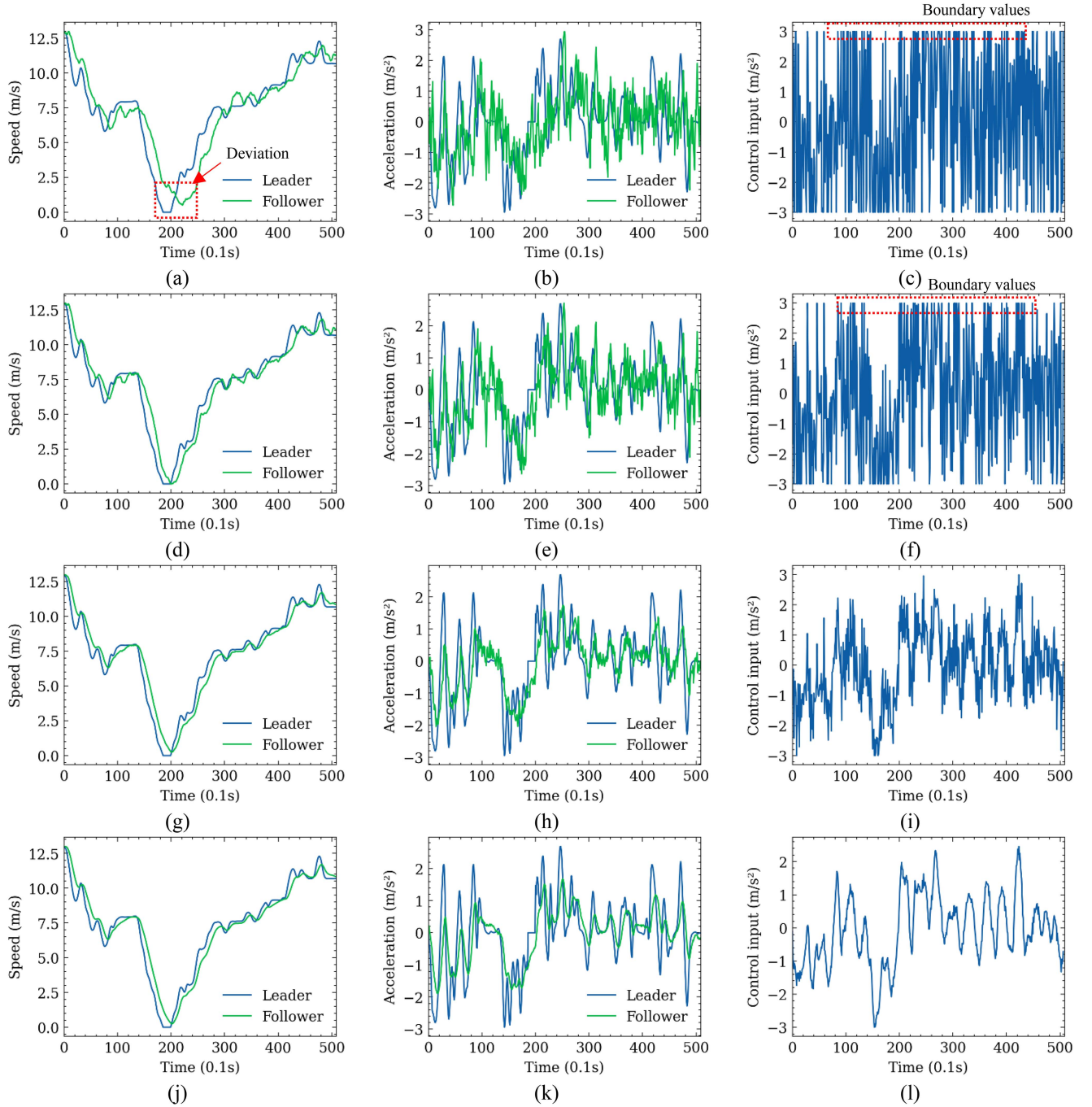


Fig. 3. Performance of ADSSRL-based strategy within a traffic oscillation: (a)–(c) Speed profiles, acceleration profiles, and control action profile after pretraining; (d)–(f) speed profiles, acceleration profiles, and control action profile after training for 10 episodes; (g)–(i) speed profiles, acceleration profiles, and control action profile after training for 100 episodes; (j)–(l) speed profiles, acceleration profiles, and control action profile after training for 1000 episodes.

traffic oscillations and the latter describes the level of driving comfort.

According to Fig. 3(a)–(c), the following vehicle has obtained elementary knowledge of car-following control. Specifically, it follows the leader to conduct the deceleration and acceleration behaviors sequentially, though not so satisfying. Statistical results show that the control efficiency and driving comfort costs are 5.84 and 0.98 m^2/s^4 , respectively. Besides, the following vehicle is exposed to traffic conflicts for 56 timesteps.

After training for ten episodes, the following vehicle moves at more approximate speeds with its leader (see Fig. 3(d)). Statistical results show that it sacrifices some driving comfort ($0.97 \text{ m}^2/\text{s}^4$) for achieving higher control efficiency (cost of 0.55) and a safer driving state (without traffic conflicts). It demonstrates that the ADSSRL-based agent promptly learns an acceptable policy via minimizing regression loss in supervised learning. However, the control input curve still has many fluctuations and boundary values (see Fig. 3(f)).

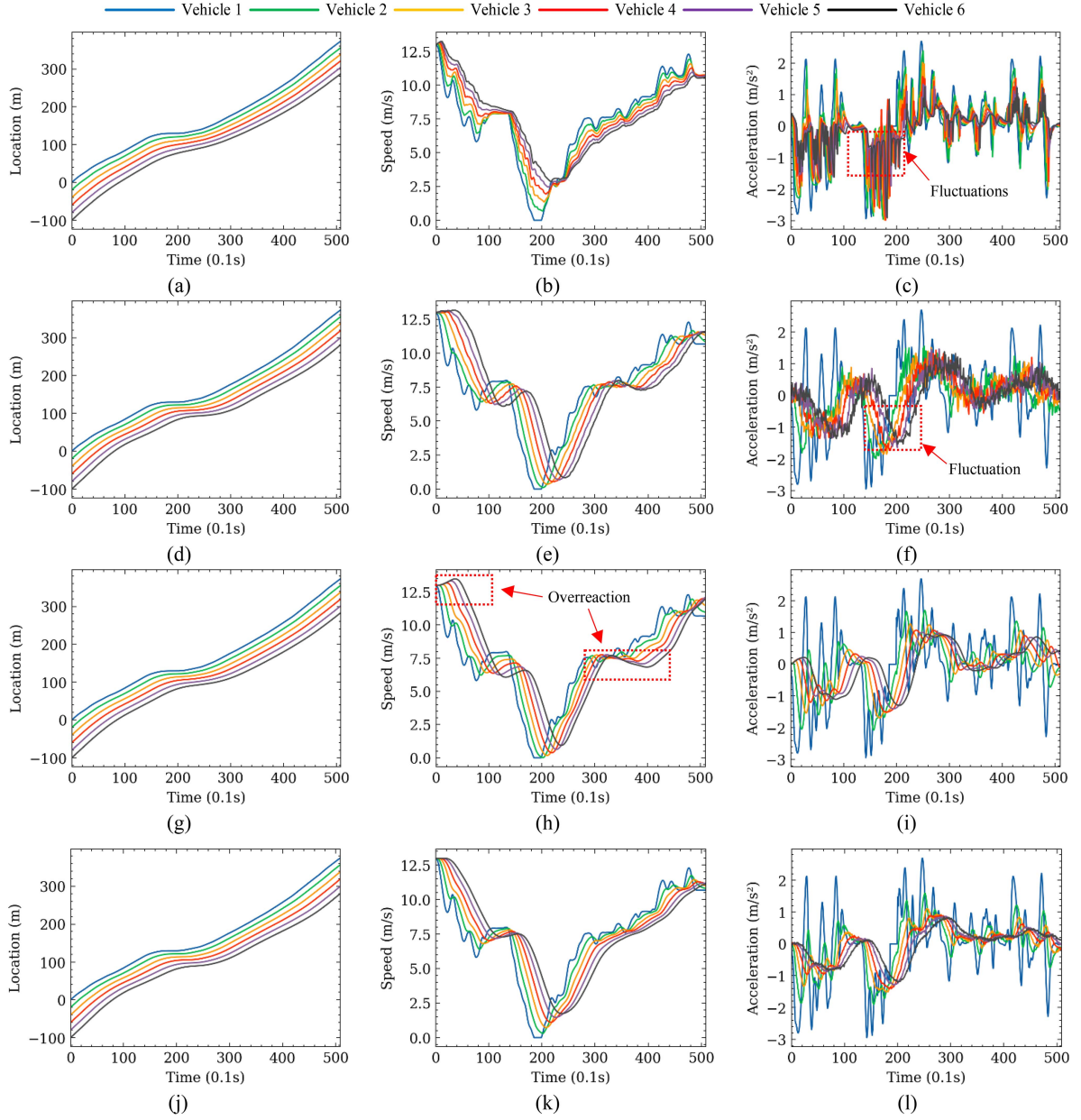


Fig. 4. Model performance for a six-vehicle platoon controlled by different strategies: (a)–(c) Vehicle trajectories, speed profiles, and acceleration profiles of vehicles with DDPG-based strategy control; (d)–(f) vehicle trajectories, speed profiles, and acceleration profiles of vehicles with PPO-based strategy control; (g)–(i) vehicle trajectories, speed profiles, and acceleration profiles of vehicles with GESOBC-based strategy control; (j)–(l) vehicle trajectories, speed profiles, and acceleration profiles of vehicles with ADSSRL-based strategy control.

According to Fig. 3(g) to (i), the ADSSRL agent achieves better car-following performance after training for 100 episodes. Only a control efficiency cost of 0.28 and a driving comfort cost of $0.64 \text{ m}^2/\text{s}^4$ are spent. At the same time, it exhibits a traffic conflict-free car-following trajectory. After training for 1000 episodes, the well-trained agents present the final training performance with smoother acceleration and control input curves in Fig. 3(k) and (l), respectively.

C. Platoon Car-Following Performance

In this section, we investigate the car-following performance within a platoon. A systematical comparison is conducted for

the performance of the proposed ADSSRL-based car-following strategy with the other three car-following strategies. A six-vehicle platoon is simulated with the first vehicle trajectory from the NGSIM data. We set the same endogenous disturbances for each vehicle i as $\delta_i(t) = 3 * \sin(0.25t)$, $i = 2, 3, \dots, 6$. After policy implementation, the position, speed, and acceleration curves are plotted in Fig. 4.

As shown in Fig. 4(b) and (c), the speed and acceleration with the DDPG-based strategy control fluctuate significantly. The acceleration curve with the PPO-based strategy control is more moderate but still has apparent fluctuations (see Fig. 4(f)). Hence, the two RL-based strategies have not converged to the expected policy.

According to Fig. 4(g) to (i), vehicles with the GESOBC-based strategy control show smoother speed and acceleration profiles. However, we can find that the followers overreact to preceding speed disturbances in some periods (see Fig. 4(h)), which decays the car-following performance. On the contrary, the curves with the proposed ADSSRL-based strategy control are more acceptable. It presents the best performance in smoothing the driving behavior and alleviating disturbances (see Fig. 4(k) and (l)).

Notably, the simulation time is set to be 50 s, which is deemed adequate for our evaluation purposes. Although vehicle platooning can occur over long distances, previous studies [50] recommend limiting the platoon size to no more than ten vehicles. We have mainly conducted simulations of a platoon consisting of six vehicles. In our experiment, we observed that within 50 s, traffic oscillation could propagate up to 450 m at an average propagation speed of 9.0 m/s through an autonomous vehicle platoon. This means that 50 s is sufficient to propagate traffic oscillation to the final vehicle of a six-vehicle platoon, as the maximum length of the platoon with a short headway (1.0 s) is only 165 m (i.e., $1.0 \times 33 \times 5$), assuming that vehicles travel at the free-flow speed of 33 m/s.

Fig. 5 demonstrates the performance of two running costs for each vehicle separately for quantitative evaluation. It is found that the proposed ADSSRL-based strategy outperforms all baselines with the least control efficiency cost and driving comfort cost. From the control perspective, the monotonically decreased running cost suggests that the proposed strategy enables vehicles to maintain close to the expected states through the vehicular string. Moreover, upstream vehicles can achieve the equilibrium consensus better due to the weakened traffic oscillations.

To conduct a more extensive comparison of model performance, we expanded the experiment to include fifty vehicles controlled by different strategies. The leading vehicle trajectory and endogenous disturbance were kept consistent with the previous experiment.

As shown in Fig. 6(b), the overall performance of our proposed policy is superior to that of the DDPG policy in terms of driving comfort, especially when the vehicle count is less than six. However, it is worth noting that the DDPG policy slightly outperforms our proposed ADSSRL policy for vehicles ranging from the 7th to 17th positions. The possible reason is that the DDPG policy sacrifices more control efficiency performance for driving comfort. Additionally, the ADSSRL policy incurs the lowest driving comfort cost (nearly zero) for the 50th vehicle. Moreover, the lowest control efficiency costs shown in Fig. 6(a) indicate that the proposed strategy consistently outperforms baselines for all vehicles.

D. Anti-Disturbance Performance

In this section, we analyze the control actions under different exogenous and endogenous disturbances to investigate the anti-disturbance mechanism. A traffic stream with 40 vehicles at the same speed of 20 m/s is simulated. The leading vehicle obeys a preset trajectory, and the following vehicles are controlled by the

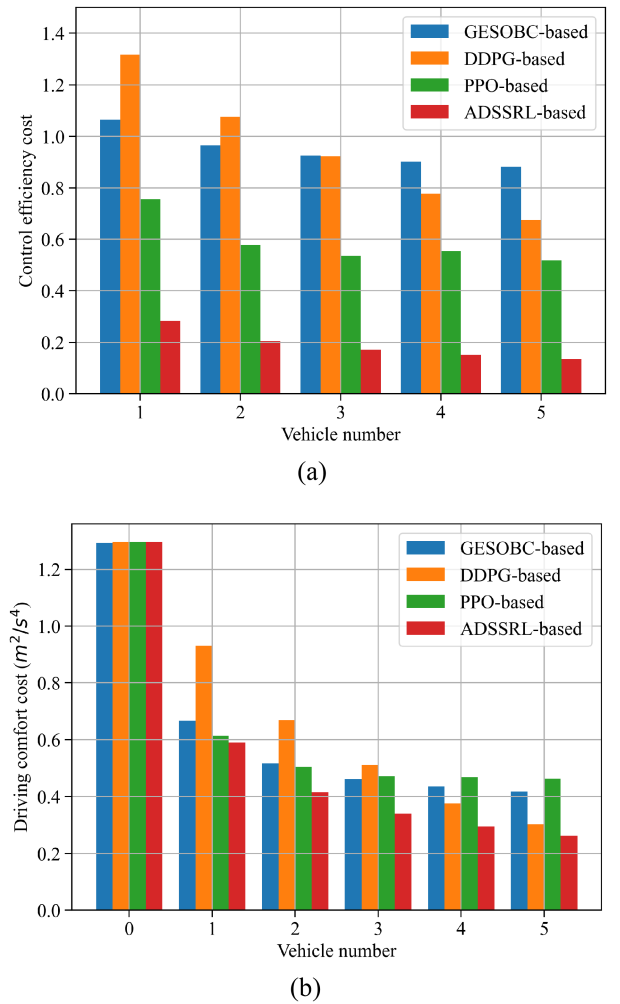


Fig. 5. Model performance for each vehicle with different strategies: (a) control efficiency cost; (b) driving comfort cost.

proposed ADSSRL-based strategy. The leading vehicle's trajectory consists of four sequential pieces: (i) keeping constant speed for 30 timesteps; (ii) decelerating to 0 m/s for 100 timesteps with -2 m/s^2 ; (iii) idling for 30 timesteps; (iv) recovering to the original speed with 2 m/s^2 and then remaining unchanged until 700 timesteps. We set different endogenous disturbances for each agent i as $\delta_i(t) = 2 * \cos(0.5 * \frac{t}{i})$, $i \in 1, 2, \dots, 39$. The synthetic trajectory with different disturbances is employed to validate the generalization capability of the ADSSRL-based strategy. It helps to demonstrate the anti-disturbance mechanism clearly.

After simulation, vehicle trajectories and speed are illustrated in Fig. 7(a)–(b). A vehicle moves at high speed with blue color and low speed with red color. Results suggest that the following vehicles extensively alleviate the preceding traffic oscillations (represented by reduced red area). In addition, the magnitude of speed reduction decreases through the vehicular string, so the followers can quickly recover from the preceding disturbances, showing outstanding robustness and resilience. Note that the frequency of endogenous disturbances in Fig. 7(c) is relatively

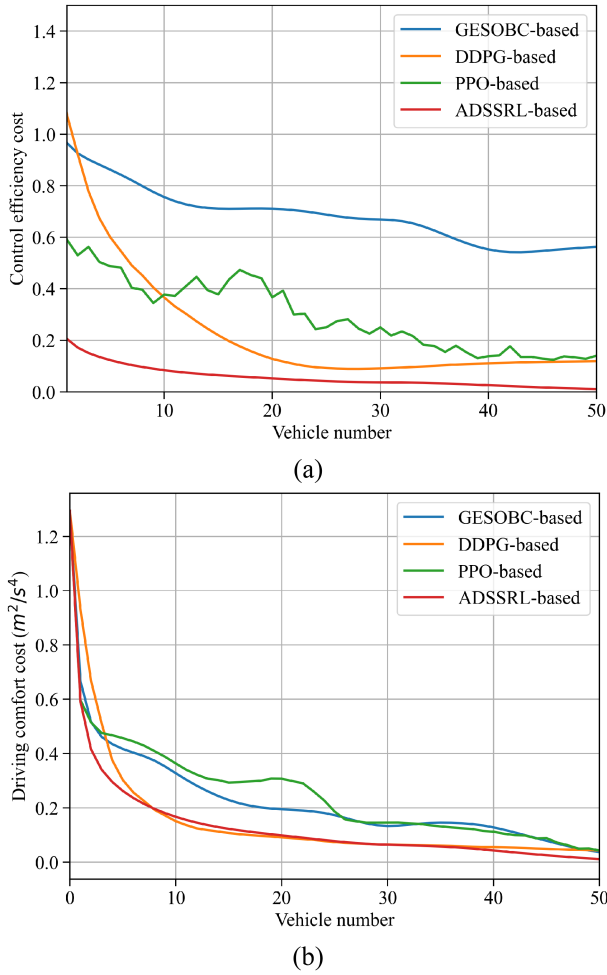


Fig. 6. Model performance with different strategies for fifty vehicles: (a) control efficiency cost; (b) driving comfort cost.

high. We designed the frequency in order to validate the proposed strategy in a harsh environment.

Now we select the 2nd, the 20th, and the 40th vehicles to exhibit the ADSSRL agent's actions (i.e., control inputs) under different disturbances, as illustrated in Fig. 7. According to Fig. 7(c)–(d), the control inputs display a similar trend as the exogenous disturbance curve. It enables the vehicle to adapt to preceding traffic oscillations. At the same time, it targets to overcome the endogenous disturbances, thereby maintaining a relatively stable acceleration profile. Hence, one can observe apparent fluctuations in the control input curve. Overall, in this case, the effects of exogenous disturbances on control inputs are larger than those of endogenous disturbances.

As shown in Fig. 7(e)–(h), it is found that the control input presents almost opposite values of the endogenous disturbances. In addition, the acceleration profiles become smoother with increased vehicle serial numbers as disturbances are gradually dampened. With the above analysis, we can summarize the anti-disturbance mechanism from two aspects: (i) attenuating the endogenous disturbances via producing approximately opposite

action values; (ii) tracking the exogenous disturbance to sustain a smoother and more moderate acceleration profile. In this way, vehicles can sequentially eliminate the continuous propagating oscillation and respective control system disturbances.

E. Effects of Endogenous Disturbance Bound

In this section, multiple field trajectories from the NGSIM dataset are selected to quantitatively evaluate the generalization capability of our proposed control strategy for different leading vehicle trajectories. We also assess the effects of endogenous disturbance bound. It is intractable to enumerate all possible disturbance cases. Without the loss of generality, we suppose three types of vehicular control system disturbances: $\delta_i(t) = D_b \sin(t)$, $\delta_i(t) = D_b \cos(t)$, and $\delta_i(t) = \min(\max(N(0,4), -D_b), D_b)$ where D_b is the selected disturbance bound and $N(0,4)$ is the Gaussian distribution with $\mu = 0$ and $\sigma^2 = 4$. We average the aggregated results for all vehicles, as shown in Fig. 8.

Typically, traffic oscillations are characterized by three primary features, namely amplitude, duration, and propagation velocity. To analyze the traffic oscillation characteristics for different policies in the experiment, we calculated the average propagation speed of traffic oscillations. Our findings indicate that the average propagation speed through controlled autonomous vehicles (9.1 m/s) is significantly higher than that through human-driven vehicles (around 5.0 m/s). The reason for this difference is that the short reaction time of automated vehicles enables them to proactively decelerate to alleviate traffic oscillations. Based on the wave propagation paths, we measured the average amplitude and duration. The results demonstrate that the ADSSRL policy presents the least average duration (14.66 s) and amplitude (5.83 m/s). These findings suggest that our proposed strategy can effectively dampen traffic oscillations, as evidenced by the minimum duration and speed reduction.

This study aims to alleviate the speed fluctuations and various endogenous disturbances, which are measured by two cost factors. First, the control efficiency cost measures the vehicle's ability to maintain the equilibrium state in car-following mode. A low control efficiency cost implies that a well-designed car-following controller can maintain harmonious speeds and equilibrium spacing when faced with a disturbance. Fig. 8(a) indicates that vehicles controlled by the proposed policy have a better ability to attenuate traffic oscillations, indicating an efficient control strategy. The driving comfort cost is the second factor, which measures the sum of the square of accelerations. Frequent acceleration and deceleration can cause uncomfortable driving experiences. A lower driving comfort cost implies relatively stable driving states with smooth speed profiles. Fig. 8(b) shows the lowest driving comfort cost values for the proposed strategy, which demonstrates its superior performance in providing a comfortable driving experience.

Statistical results show that the car-following system with ADSSRL-based strategy control spends the least running cost in all cases. It achieves at least a 44.5% reduction in control efficiency cost and a 10.1% reduction in driving comfort cost

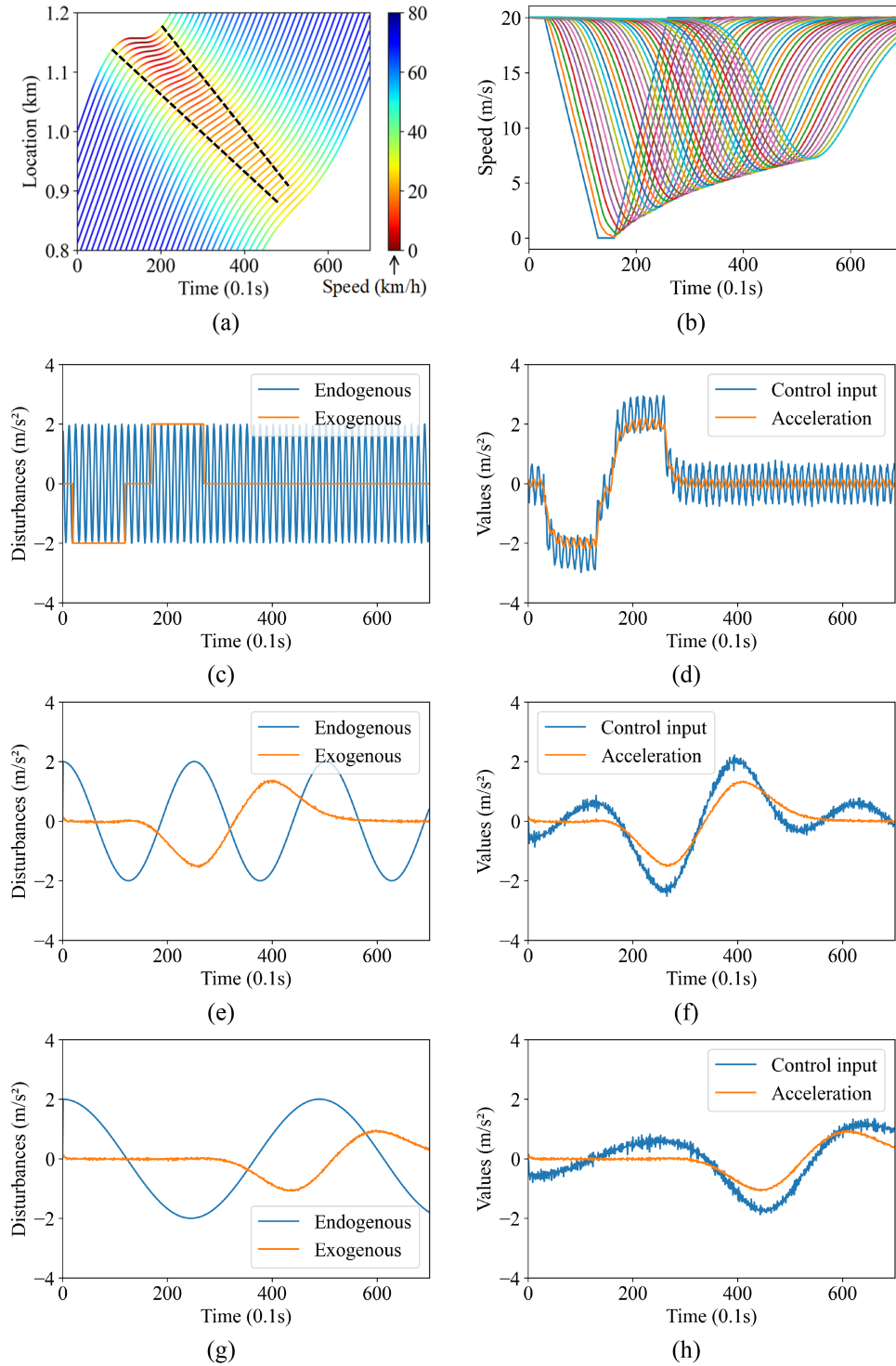


Fig. 7. Model performance for a vehicle stream within a traffic oscillation: (a) Vehicle trajectories; (b) speed profiles; (c)–(d) disturbances, control inputs and accelerations of 2nd vehicle; (e)–(f) disturbances, control inputs and accelerations of 20th vehicle; (g)–(h) disturbances, control inputs and accelerations of 40th vehicle.

compared with all baselines when $D_b = 4$. We also find that, unlike other strategies, the proposed strategy is insensitive to the disturbance bound. One can observe that the GESOBC-based controller performs better than the DDPG-based strategy in all

cases. The reason might be that DDPG cannot learn an effective policy from the environment with misguided reward feedback. By contrast, the GESOBC-based strategy has a relatively robust performance under different disturbances.

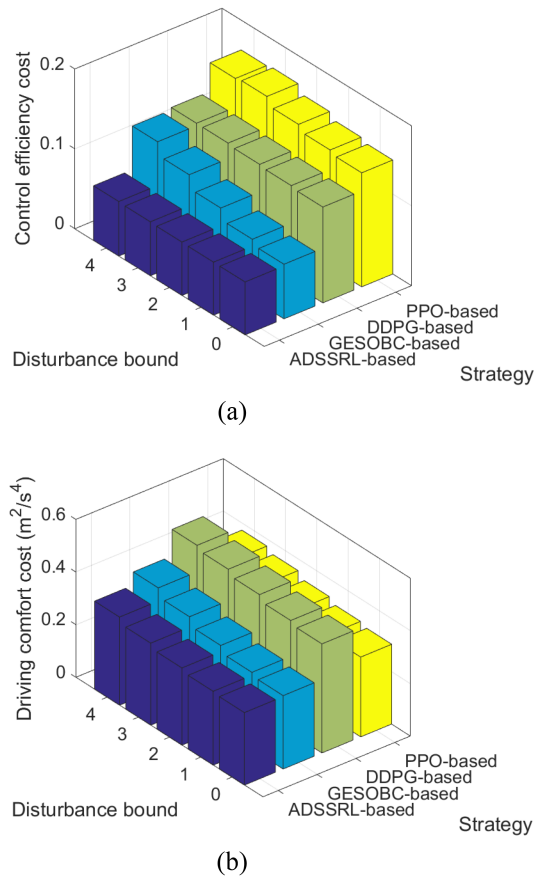


Fig. 8. Model performance in different disturbances: (a) control efficiency cost; (b) driving comfort cost.

V. CONCLUSION AND DISCUSSION

This study proposed a car-following model for overcoming random disturbances originating from traffic oscillations and vehicular control systems. To facilitate this, we modified a robust controller to generate an idealistic expert demonstration. Guided by the demonstration, the RL-based car-following policy is trained in a random traffic environment. The robust expert can accelerate RL training and help develop the ability to attenuate disturbances. The self-supervised RL algorithm is employed to enable the training framework and drive stable RL policy improvement in a perturbed car-following environment.

After training, the ADSSRL-based strategy presented a better convergence performance than other RL algorithms, manifesting as a smooth and almost monotonously increasing reward curve. It gained the most significant final reward eventually. A series of experiments were conducted to validate the proposed strategy. Multiple field trajectories from the NGSIM dataset were selected to evaluate its generalization capability quantitatively. It achieved at least a 44.5% reduction in control efficiency cost and a 10.1% reduction in driving comfort cost compared with three baselines given a disturbance bound of 4 m/s². We calculated the two indexes for each vehicle separately within a platoon. They decreased through the vehicular string. It demonstrated the effectiveness of the proposed strategy in alleviating traffic oscillations and controlling system disturbances. We have conducted

tests to evaluate the performance of the proposed algorithms in terms of repeatability under varying environmental conditions. The environment is subject to both endogenous disturbances containing parameter uncertainty and exogenous disturbances containing acceleration noise from ground-truth data. The model performance controlled by our proposed ADSSRL policy is found to be relatively stable. This stability can be attributed to the application of RL with expert preview to guide policy training and develop a higher-quality control strategy.

This study applies human knowledge (a well-designed robust controller) to guide RL training, thereby developing a higher-quality control strategy. A stable and efficient RL training approach is successfully applied in autonomous vehicle control with disturbances. We employ a random sampling method to extract 512 state-action data points from a replay buffer with a size of 2000. In future research, we will delve further into more intricate sampling methods, such as developing a hierarchical sampling system to effectively select various state-action pairs. In addition, we can develop more ‘safety net’ approaches to achieve contingency checks. For example, we can set safety constraints in the training stage to ensure that the RL algorithm does not learn any behavior that violates safety rules. We can also implement human intervention to take over control of the vehicle in case of emergencies. It continuously monitors the performance of the RL algorithm in real-world conditions and retrain the algorithm periodically to improve its performance.

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